varquanMS: local-scale, adaptive, variable-quantization-length compression method for radio interferometry data

Abstract

We present varquanMS, an experimental archive for radio-interferometric Measurement Sets using local scaling and adaptive integer precision. Each baseline–frequency block stores a complex median, range, power-spacing parameter and a selected component width. A dimensionless epsilon-D criterion bounds reconstruction error relative to local component median deviation, with maximum, bias and baseline guards. Nine widths from 3 to 16 bits plus exact float32 fit within a four-bit selector. Original UVW, sampling and retained metadata are preserved. On one calibrated 15-integration OVRO-LWA observation, epsilon-D 0.01, 0.04 and 0.18 yield visibility payloads of 22.859, 17.433 and 9.932% of logical complex64 bytes. At 0.18, a complete archive is 242.769 MB versus 985.236 MB for the same-policy retained-data lossless archive. The calibrated profile explicitly discards raw/model columns and flagged values while retaining flags; its four-bit weights are exact on this dataset. Additional I/V imaging above 10° gives difference RMS of 4.810–4.876% and 10.056% of the matching reference residual RMS, respectively. A follow-up extends V validation to all three epsilon values. Three repeated full-MS conversion runs reduce median time from 485.870 to 72.685 s (6.685×), including verification; the 15-second target remains unmet. These measurements establish agreement with the supplied calibrated reference, not sky truth or general polarization fidelity.

1. Introduction

An interferometer records complex correlations for many antenna pairs, frequency channels and times. The visibility array grows approximately as $N_{\mathrm{ant}}^2 N_\nu N_t N_p$, making storage and repeated data transfer costly. Complex64 uses two float32 components: eight bytes per visibility. Averaging can reduce volume, but also changes time or frequency resolution. We investigate lossy precision reduction while retaining the original sampling and row identities. The intended output is an independently decodable archive from which a new MS can be materialized for existing imaging software.

A Measurement Set includes coordinates, array-valued columns, descriptors, keywords and references to subtables. Correct archival reconstruction must account for this structure. Casacore also separates logical columns from physical storage managers, so repeated logical values do not by themselves establish wasted disk space. We report allocated payload and complete-file bytes separately. MS v2 specification, Casacore table system.

Normalization, nonlinear quantization and dithering are established techniques in visibility compression. Dysco combines these methods, including antenna/frequency normalizations, and evaluates their effects on LOFAR and MWA observations. Its published results motivate direct image-domain validation, but do not predict fidelity for the calibrated observation studied here. The present work investigates local precision selection, explicit exceptions, a dimensionless error rule and MS reconstruction. It does not claim normalization or nonlinear quantization as new, and a matched Dysco comparison has not yet been performed. Offringa (2016).

Array-based schemas such as XRADIO/MS v4 provide a useful separation between scientific coordinates and storage representation; this prototype exports MS v2 and does not claim MS v4 conformance. XRADIO overview. Blockwise low-bit formats and scale compression also appear in machine-learning quantization, including QLoRA. Here, all scale factors remain

float32, no learned model is required, and model-quality results provide no scientific-fidelity assurance for visibilities. Dettmers et al. (2023).

2. Method

2.1 Inspection, partitioning and grouping

The input is inspected before encoding. The implemented regular path requires complex64 visibility cells with channels inside each cell and a homogeneous configuration within each partition. Multiple contiguous integrations are handled as separate partitions. Unsupported layouts fail explicitly. Ordered antenna pairs and distinguishing MS row context are preserved; antennas are not swapped and visibilities are not conjugated to canonicalize a baseline.

The internal visibility axes are baseline/row, channel and correlation. Product names and order are read from POLARIZATION metadata. Autocorrelation and cross-correlation rows are separated. For each product p, a stable amplitude ordering uses

$$A_{b,p} = \mathrm{median}_{\nu \in \mathcal{V}_{b,p}} \mid V_{b,\nu,p} \mid,$$

where $\mathcal{V}$ excludes flagged and nonfinite samples. Original row index breaks ties, and rows with no eligible values form a deterministic special group. Each baseline spectrum stays intact. The stored permutation obeys

$$V^{\mathrm{archive}}_{j,\nu,p} = V^{\mathrm{source}}_{\pi_p(j),\nu,p}, \qquad V^{\mathrm{restored}}_{\pi_p(j),\nu,p} = \hat{V}^{\mathrm{archive}}_{j,\nu,p}.$$

A quantization block contains at most B consecutive archived baselines and F consecutive channels for one product. Exact edge bounds and class boundaries are stored. The present experiment fixes $B = 128$ and $F = 48$, independently per product. No time/channel averaging is performed. Sorting changes which values share factors; the permutation itself removes no information and has a measured byte cost. The implementation also supports original, baseline-length, orientation, antenna and amplitude-bucket orders, but this three-tolerance experiment does not compare those alternatives.

Figure 1a illustrates the grouping operation: rows move as whole spectra, while channel order stays fixed. Each outlined block shares a complex centre, range, power-spacing parameter and selected component width. Figure 1b connects those groups to the reconstruction-error checks and precision decision.

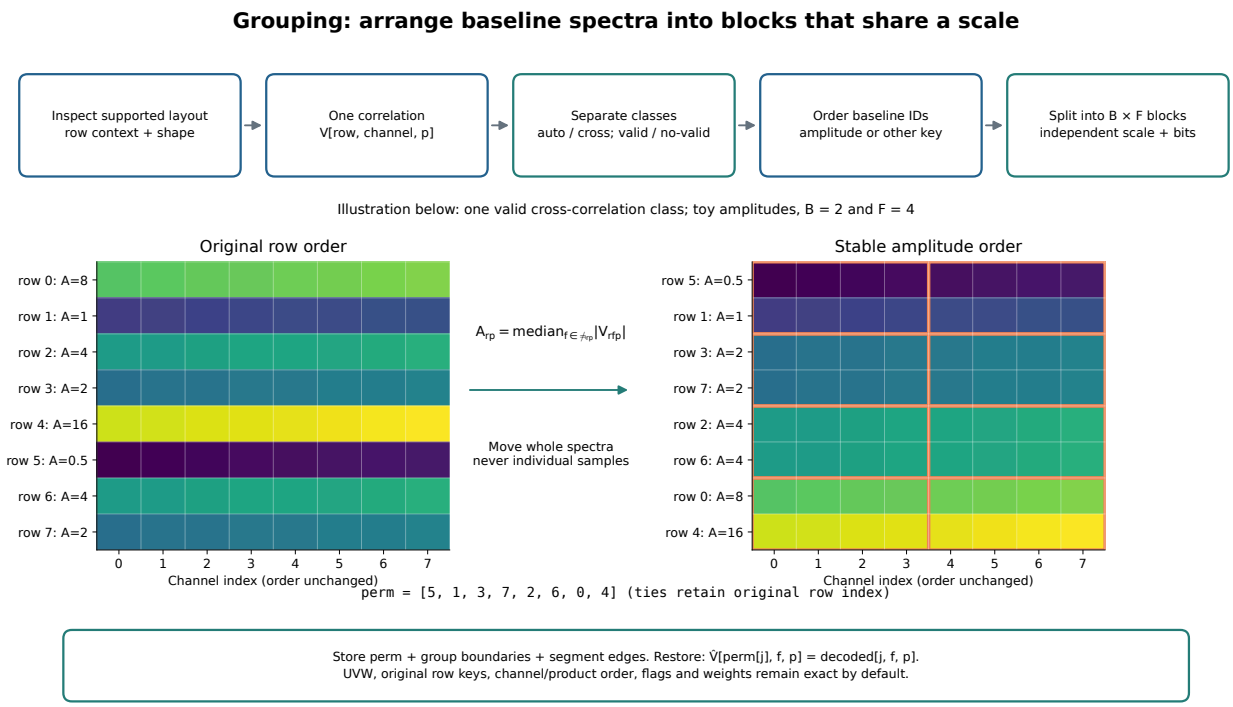


Figure 1a. Grouping concept for one integration, configuration and correlation product, after separating auto/cross classes. The illustrative amplitudes show stable whole-spectrum sorting; equal amplitudes retain original row order. Orange outlines mark blocks of two baselines by four channels. This small schematic illustrates segmentation; the measured calibrated-data experiments use 128 baselines by all 48 channels per product, with partial edge blocks. The stored permutation restores original row positions without channel averaging or antenna-pair reversal. Colours illustrate amplitude, not measured data.

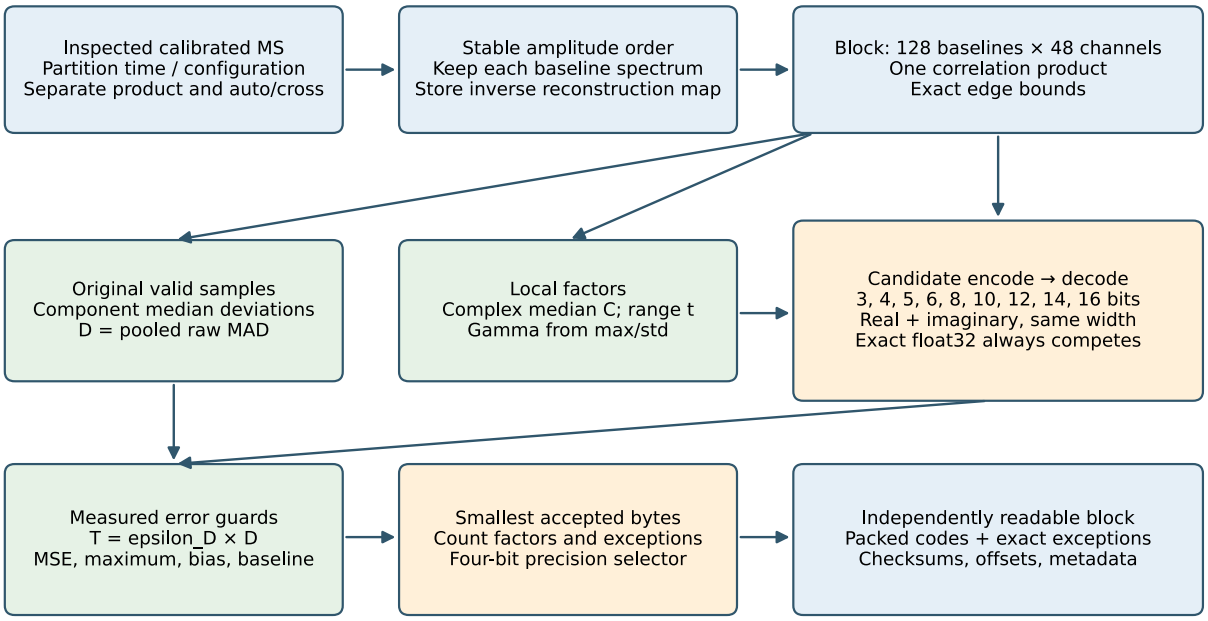


Figure 1b. Spectra are ordered within one configuration, integration, product and auto/cross class. Each block is independently encoded, reconstructed and checked; the smallest accepted serialized candidate is stored with its reconstruction map.

2.2 Local centre, range and power spacing

For block q, store two component medians $C_q = C_{q,R} + iC_{q,I}$, a shared range $t_q \geq 0$ and a power parameter γ_q. The encoder computes the medians from eligible values, excluding flags, nonfinite values and protected signed-zero samples, then rounds the medians to float32. Residuals are calculated against these stored medians. Two range candidates are tested: the maximum and the 99.5th percentile of $\max(|\,\Re(V - C_q)\,|, |\,\Im(V - C_q)\,|)$. Each range is rounded upward to float32. Values outside a candidate range are preserved through exact exceptions.

For either real component x, define $y = (x - C)/t_q$ on $[-1, 1]$. Let

$$M_b = 2^b - 1, \qquad u_k = \frac{2k - M_b}{M_b}, \qquad k \in \{0, ..., M_b\}.$$

The reconstruction levels are

$$\hat{x}_k = C + t_q \,\mathrm{sgn}\,(u_k)\;|\;u_k\;|^{1/\gamma_q}\;.$$

All 2^b unsigned codes are used. The residual grid is symmetric and spans $[-t_q, t_q]$, without a separate sign bit or a duplicate signed-zero code. Since M_b is odd, the grid has no exact residual-zero level for $t_q > 0$. This is a midrise grid: an ordinary value at the centre can acquire a small error. Negative zeros are protected exactly through exceptions. If $t_q = 0$, the decoder copies the stored component centres directly. Entirely zero or otherwise compressible blocks may instead select the exact representation.

For $\gamma_q = 1$, the reconstruction spacing is uniform, $\Delta_b = 2t_q/(2^b - 1)$. For $0 < \gamma_q < 1$, levels concentrate near the centre:

$$\left|\frac{d\hat{x}}{du}\right| = \frac{t_q}{\gamma_q}\;|\;u\;|^{1/\gamma_q - 1}\;.$$

The candidate encoder first computes the continuous coordinate

$$k^* = \frac{M_b}{2}[1 + \operatorname{sgn}(y) \mid y \mid^{\gamma_q}].$$

It reconstructs the adjacent integer codes $\lfloor k^* \rfloor$ and $\lceil k^* \rceil$ using the stored factors and chooses the closer reconstructed float32 value. Equal errors select the even code. Choosing by reconstructed values matters for nonlinear spacing; ordinary rounding of k^* alone need not choose the closest level in data units. Unflagged outliers are never silently saturated. Endpoint clipping only protects the integer lookup from floating-point roundoff after the range/exception decision.

The tested gamma rule uses the maximum absolute component residual divided by pooled within-component population standard deviation, r_q. Both statistics use all eligible residuals before selecting range exceptions, with float64 arithmetic and variance normalization `ddof=0`. It selects

$$\gamma_q = \begin{cases} 1, & r_q \le 4 \text{ or the ratio is undefined,} \\ 1/2, & 4 < r_q \le 8, \\ 1/4, & r_q > 8. \end{cases}$$

This is a deterministic encoder heuristic, not a noise model. The selected gamma is stored, so decoding never estimates it again. Full gamma search is available as a separate experiment. The decoder evaluates normalized codes and powers in float64, using explicit squares for gamma $1/2$ and fourth powers for gamma $1/4$, then multiplies by the stored range, adds the stored centre and rounds once to float32. The reference implementation fixes the operation order.

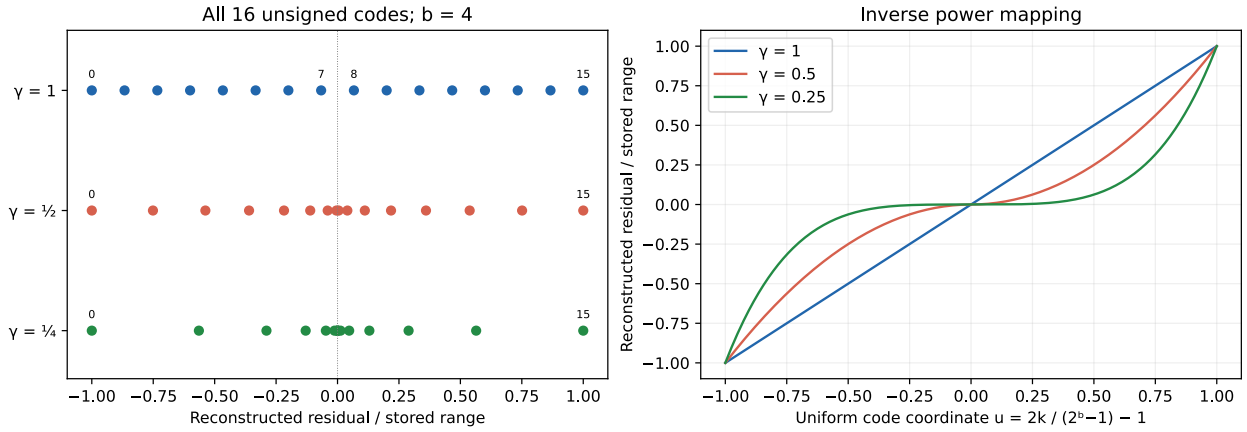


Figure 2. Illustrative four-bit reconstruction levels after median subtraction. Every row contains 16 distinct unsigned codes. Smaller gamma increases resolution near the centre while increasing spacing near the range endpoints. The experiment also allows eight other component widths.

2.3 Choosing precision with epsilon D

The reference statistic for acceptance is computed from the original finite, unflagged samples, before outlier selection. Define raw component deviations

$$d_{q,R} = \operatorname{median} \mid \Re V - \operatorname{median}(\Re V) \mid, \qquad d_{q,I} = \operatorname{median} \mid \Im V - \operatorname{median}(\Im V) \mid,$$

$$D_q = \sqrt{\frac{d_{q,R}^2 + d_{q,I}^2}{2}}, \qquad T_q = \epsilon_D D_q.$$

These medians use float64 arithmetic and no Gaussian conversion factor. D_q describes signal variation; it is not a thermal-noise estimate. It is separate from the rounded median used

for coding. Multiplying all visibilities by a real amplitude factor scales D_q and the permitted absolute error by its magnitude, subject to finite-precision effects. Thus the tolerance has no fixed amplitude unit and requires no denominator floor near an individual zero visibility.

For reconstructed error $e = \hat{V} - V$, the tests are

$$\frac{1}{2N_q}\sum_{n\in\mathcal{V}_q} |\, e_n \,|^2 \le T_q^2, \qquad \max_{n\in\mathcal{V}_q} |\, e_n \,| \le 10T_q,$$

$$\left|\frac{1}{N_q}\sum_{n\in\mathcal{V}_q} e_n\right| \le T_q, \qquad \max_{b\in q}\frac{1}{2N_{q,b}}\sum_{n\in\mathcal{V}_{q,b}} |\, e_n \,|^2 \le 2T_q^2.$$

Only baselines with eligible samples contribute to the last expression. These are pooled real/imaginary constraints; separate component errors are measured but there is no independent per-component relative ceiling. A zero D_q imposes zero measured error, even when a sparse nonconstant block has zero MAD. Empty valid sets have zero measured errors. Exact fallback is always available.

The precision ladder is

$$b \in \{3, 4, 5, 6, 8, 10, 12, 14, 16\} \cup \{\text{exact float32}\}.$$

Each bit width and range candidate is actually reconstructed and tested. The encoder chooses the candidate with the fewest serialized bytes, including the header, codes, exception stream and optional lossless compression. Exact storage also competes on size and wins equal-size ties. A larger epsilon generally permits coarser precision, but bytes and image error are measured rather than inferred from the nominal width. Precision is common to the real and imaginary components within a block. A b-bit component codec uses $2b$ payload bits per complex sample before exceptions and other overhead.

The general interface also supports sampled global std, block std and uncentered block RMS references through ϵ_G, ϵ_S and ϵ_R. Those coefficients are disabled in the present experiment. Their sampling and combination study is documented in the 16% study; no fixed absolute-error threshold is active here.

2.4 Data format and precision tags

The scientific schema is independent of its HDF5 prototype container. The archive holds typed table descriptors and keywords, exact supported subtable values, original coordinates, row/cell maps, permutations, flags, weights, visibility streams, error summaries and integrity inventories. Shared subtable content is stored once when compacting the integration partitions. Explicit little-endian dtypes and JSON metadata avoid executable object serialization.

Each block has a 16-byte base header containing a precision selector, compression identifier, mapping identifier, complex sample count, exception count and a float32 range or step. Power blocks append two float32 centre components and float32 gamma, making 28 bytes before the body. An exact block uses the 16-byte base header and original complex64 bits. The first byte of the new header is `0xA0 | tag`, with the lower four bits interpreted as follows:

Tag	Interpretation
0	Exact complex64
1–9	3, 4, 5, 6, 8, 10, 12, 14, 16 bits/component, respectively
A	Legacy two-bit representation, outside the tested ladder
B–E	Reserved; decoder rejects
F	Invalid/error; never a successful encoded block

Four bits suffice to select the precision, not to represent every visibility. The upper nibble identifies this header version; aligned header sizes remain 16/28 bytes. The format does

not claim an additional half-byte saving per block. Historical literal-width headers remain decodable, while unsupported tags fail explicitly.

Codes are interleaved real then imaginary and packed least-significant bit first. Each width occupies its actual number of bits; 14-bit codes can cross three bytes. Unused final high bits must be zero. Exact exceptions store increasing uint32 sample positions and original eight-byte complex64 values. They restore protected special-value bits and outliers after ordinary reconstruction. Block bodies use DEFLATE only when it reduces their size. Offset indexes and per-block CRC32 allow independent reads; dataset SHA-256, container checksums and a manifest inventory provide further corruption detection. These checks detect accidental damage; they are not authentication signatures. The decoder needs no remote service, learned model or downloaded codebook. A standard-library reference block decoder and small literal conformance fixtures accompany the implementation.

2.5 Compression, decompression and calibrated-only policy

Compression proceeds through inspection, context partitioning, grouping, candidate selection, payload writing and source-aware verification. Partial outputs remain distinguishable from completed archives. The source MS is never overwritten. Decompression checks inventories and block bounds, reconstructs values, applies exact exceptions and inverse permutations, rebuilds subtable references, and writes a new MS. Original UVW float64 values are retained in metres; no antenna position, time or frequency calculation regenerates them. Bitwise-identical coordinates may be deduplicated, without tolerance-based merging.

The calibrated-only profile explicitly retains CORRECTED_DATA and removes the original DATA and MODEL_DATA. Original flagged visibility values are discarded and represented by positive-zero placeholders, while FLAG, FLAG_ROW and FLAG_CATEGORY remain preserved. It retains sample presence, array shapes and resolution. Consequently this profile is not an exact round trip of the original MS, even when the retained visibilities choose exact storage. On export, DATA is recreated from decoded CORRECTED_DATA for MS compatibility; original raw DATA is not recovered. Remaining supported metadata and auxiliary columns stay exact.

Weights use an independent four-bit codebook with checked float32 fallback; weight approximation, if needed, follows its separately recorded tolerance. The weights of this particular observation are reconstructed bitwise exactly. No quantization error estimate overwrites original weights. Optional removal of flags is a distinct destructive profile and is not enabled in these experiments.

For the supported multi-integration case, the reference workflow is:

```
export PYTHONPATH=src
python scripts/timeslot_archive.py encode INPUT.ms blocks.bundle \
  --config configs/calibrated-epsd-018.json --report encode.json
python scripts/timeslot_archive.py compact blocks.bundle archive.h5 \
  --report compact.json
python scripts/timeslot_archive.py decode archive.h5 RESTORED.ms \
  --report decode.json
```

All outputs must be new paths. Decoding receives the archive alone. Source-aware validation is a separate step and does not supply missing reconstruction data.

3. Validation on calibrated OVRO-LWA data

3.1 Observation and experiment design

The inspected input `lwa-15slot-flat.ms` contains 15 contiguous integrations, 931,920 rows and 352 antennas. Each integration has 62,128 ordered antenna pairs: 352 autocorrelations and 61,776 cross-correlations. Each cell has 48 channels and four products in metadata order XX, XY, YX, YY. Channel centres range from 63.965576 to 68.463623 MHz at 95.703125 kHz

spacing. Original TIME and TIME_CENTROID values and their UTC references are retained. The table contains 926,641 bitwise-distinct original UVW vectors.

CORRECTED_DATA contains 178,928,640 complex samples, or 1,431,429,120 logical bytes. There are 41,752,612 flagged samples. The complete source occupies 4,428,203,949 bytes, including raw and model columns that the calibrated profile deliberately omits. These different denominators must not be conflated. Source HISTORY records CASA `applycal`; this establishes the supplied calibration history, not an independent validation of the calibration or flux scale. Input inspection.

We fix all grouping, median, gamma, exception and preservation settings and test $\epsilon_D = 0.01$, 0.04 and 0.18 with the expanded nine-width ladder. The coefficients are specified in advance and are not retuned to a byte target or selected from image outcomes. Each case reads the original calibrated source; none recompresses a previous quantized result. All use 128×48×1 blocks, independent amplitude ordering, the max/std gamma rule, range quantiles 1 and 0.995, and exact fallback. The initial three-epsilon study used 24 encoder workers. A subsequent speed and I/V study retains the epsilon-D 0.18 scientific settings and decoded values, with a different execution configuration (Section 3.6). Exact configurations and versioned reports accompany both studies.

3.2 Reconstruction and imaging checks

The validation suite exercises every width, partial blocks, special values, flags, signed zeros, exact fallback, malformed/truncated streams and both header representations. Independent standard-library, portable NumPy and compiled reconstructions are compared. Native modules built before a requested width fall back to portable encoding. The subsequent performance validation passes 124 tests. Source-aware archive/export checks compare original supported metadata, coordinates, flags and weights exactly, and independently recompute normalized visibility error criteria. Source checksums are checked against the original import inventory.

Imaging uses WSClean 3.7 in the same recorded container for reference and quantized MSs. A private MS copy protects each input from imaging side effects. The initial study uses Stokes I; the subsequent epsilon-D 0.18 study runs Stokes I and V separately against matching references. A further V sweep images the same three epsilon cases, each twice, using their verified exports. All use 2048×2048 pixels, four-arcminute pixels, Briggs robust 0 and multiscale CLEAN. The remaining settings are eight threads, scale bias 0.8, 150,000 iterations, mgain 0.95, auto-threshold 0.5, auto-mask 3, local RMS, a 10° horizon mask, 40% memory setting, reordering and no model-column update. Candidate images use the reference restoring beam. Auto-masks are estimated independently, so deconvolution-path changes contribute to the measured differences. WSClean is an established wide-field imager; these particular validation settings and results belong to this experiment. Offringa et al. (2014).

The comparison mask requires elevation above 10° for all 15 integrations and contains 2,245,726 pixels. For $S \in \{I, V\}$, let S_0 be the restored reference, S_ϵ the corresponding quantized image, and $R_{0,S}$ the reference residual map. We report uncentered, unclipped statistics on this fixed mask:

$$E_S = \mathrm{RMS}_{\mathcal{M}}(S_\epsilon - S_0), \qquad \eta_S = 100\frac{E_S}{\mathrm{RMS}_{\mathcal{M}}(R_{0,S})}.$$

The I and V reference residual RMS values are 1.031244811 and 0.384439444 Jy/beam, respectively. Each denominator is a single scalar measured from the matching reference residual; there is no pixelwise residual division. These percentages are not fractional flux loss or thermal-noise budgets. Residual maps contain sky and deconvolution structure. We additionally compare dirty images, faint-region RMS, reference-peak changes and visibility

errors by baseline, product and frequency. Faint pixels are defined by reference intensity below 1% of its peak. Each quantized imaging case is CLEANed twice, and the I/V study includes a repeat of each uncompressed reference. Two runs measure observed variation, not a statistical confidence interval.

3.3 Three-epsilon Stokes I comparison

εD	Payload MB (% of complex64)	Complete archive MB	Difference mJy/beam, first / repeat	Difference / reference residual RMS %, first / repeat
0.01	327.207 (22.859%)	427.429	12.9866 / 12.9944	1.2593 / 1.2601
0.04	249.542 (17.433%)	350.112	11.8731 / 11.8666	1.1513 / 1.1507
0.18	142.163 (9.932%)	242.766	57.5807 / 49.4859	5.5836 / 4.7987

εD	Dirty difference mJy/beam	Worst reference-peak change %	Worst faint-region RMS difference %	Candidate residual RMS Jy/beam, first / repeat
0.01	1.5080	0.00020	1.0430	1.030953 / 1.030953
0.04	5.4572	0.00244	0.9556	1.031218 / 1.031221
0.18	40.0807	0.43340	4.5891	1.026183 / 1.028690

εD	Cross XX RMSE / signal RMS %	Cross XY %	Cross YX %	Cross YY %
0.01	0.59372	0.38169	0.48786	0.45558
0.04	2.35604	1.43210	1.87325	1.78090
0.18	11.49117	10.65014	11.56520	9.94197

Visibility ratios divide component reconstruction RMSE by the original component signal RMS, pooling all finite unflagged cross-correlations in each product. They are distinct from the local MAD-normalized acceptance criterion. Cross-hand visibility statistics do not constitute a polarized-image validation.

All three archives passed full-source export verification. Original supported metadata, UVW, flags and this observation’s weights passed exact checks; all 108 source-file hashes still match. All six CLEAN runs reached the final threshold, with 3 major iterations each. The new 5/10/14-bit choices encode 26,333,664, 104,509,248 and 0 complex samples, respectively, summed across the three separate experiments. Exact blocks often represent highly compressible zeros, so sample fractions and byte fractions differ. Complete distributions and auto/cross splits are retained in the machine-readable experiment report.

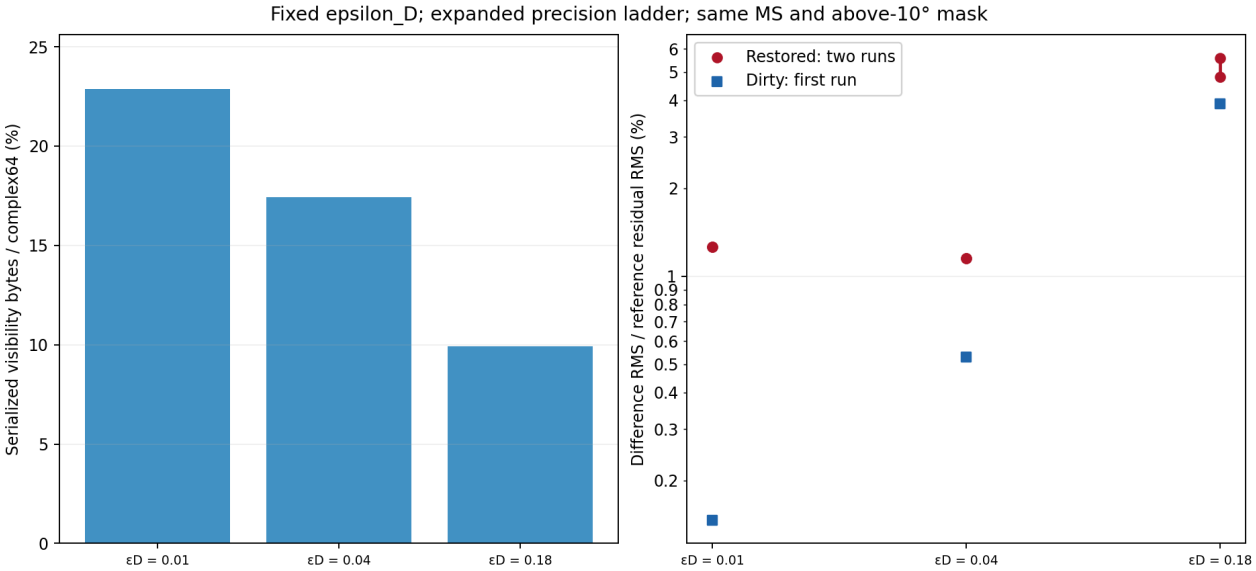


Figure 3. Fixed epsilon-D values produce measured sizes and image errors; there is no imposed byte target. Red points show two CLEAN runs. Blue squares show the first dirty-image difference, all normalized by the same reference residual RMS.

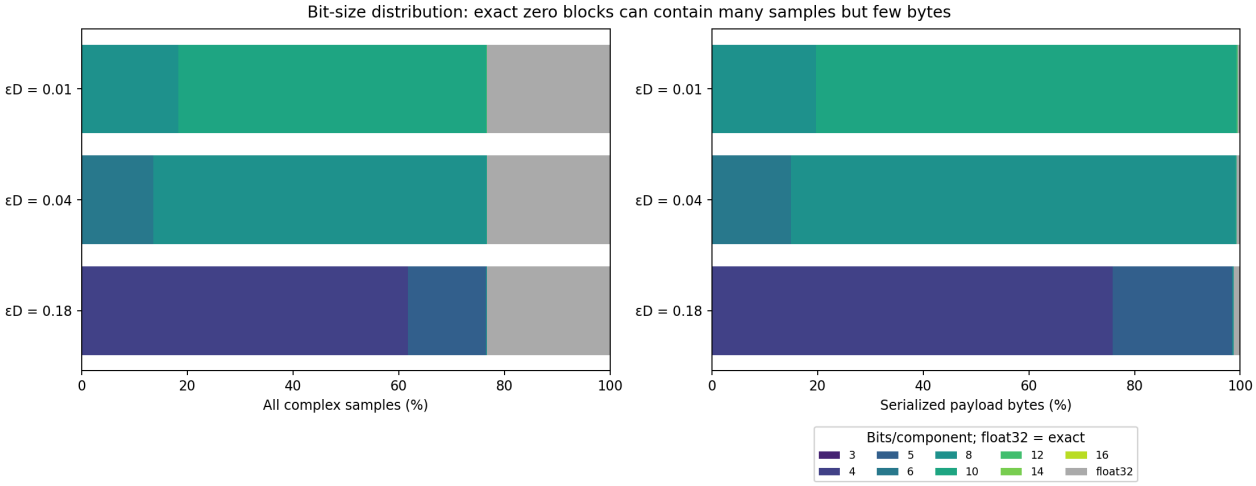


Figure 4. Selected component widths by number of complex samples and by serialized payload bytes. Float32 denotes exact storage, including cheaply compressed zero blocks. Metadata outside visibility blocks is accounted for separately.

For the pixelwise I and V displays in Figures 5–8, define

$$P_{\epsilon,S}(\boldsymbol{\theta}) = 100\,\frac{S_\epsilon(\boldsymbol{\theta}) - S_0(\boldsymbol{\theta})}{\mathrm{RMS}_{\mathrm{residual},S}}, \qquad \mathrm{RMS}_{\mathrm{residual},S} = \sqrt{\frac{1}{N_{\mathcal{M}}}\sum_{\boldsymbol{\theta}\in\mathcal{M}} R_{0,S}(\boldsymbol{\theta})^2}, \quad S \in \{I, V\}.$$

The denominators are the uncentred reference residual RMS over the full above-10° mask: 1.031244811 Jy/beam for I and 0.384439444 Jy/beam for V. Each remains fixed across epsilon cases and zooms, matching the normalization of the tabulated image-error metrics. Colourbar ticks express the ratios as percentages, with the compact label $(S_{\mathrm{quant}} - S_{\mathrm{ref}})/\mathrm{RMS}_{\mathrm{residual}}$.

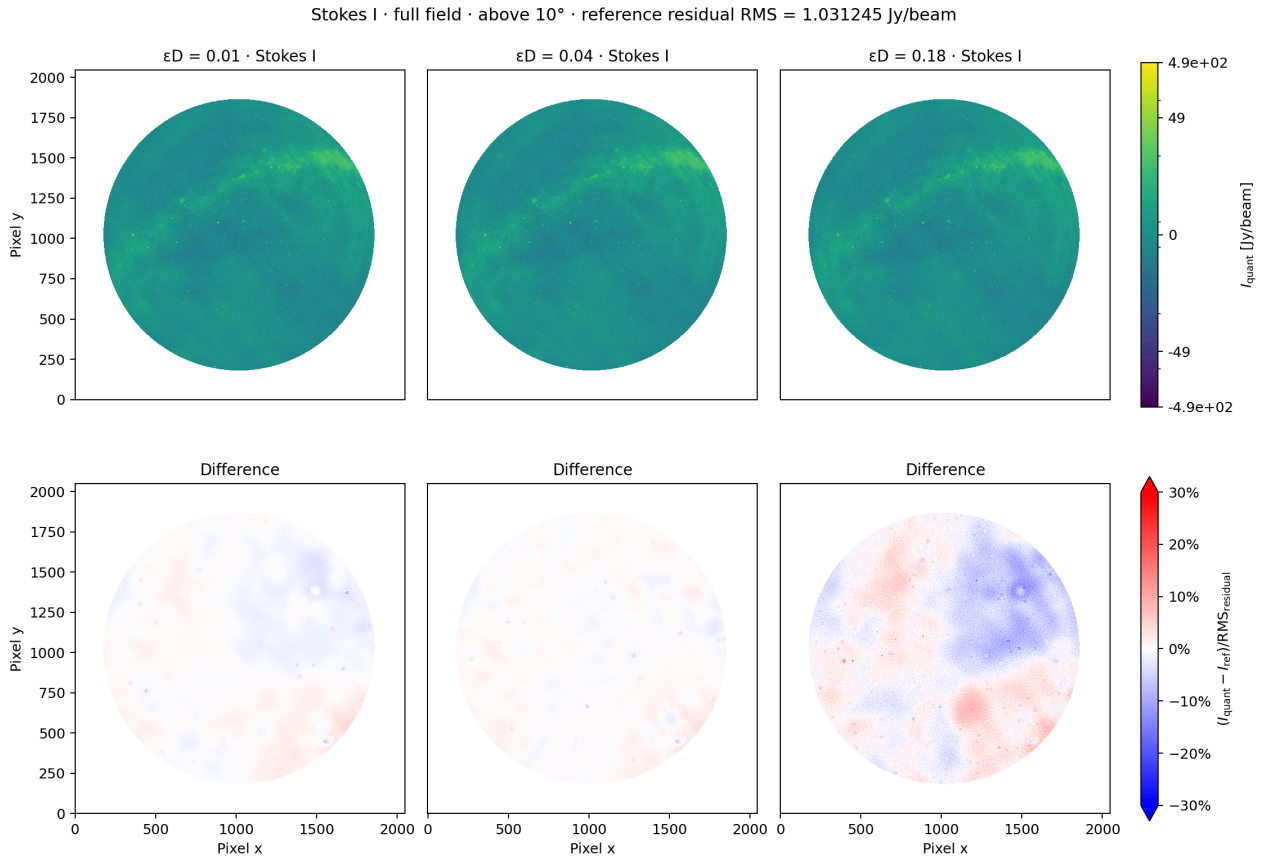


Figure 5. Two rows and three columns, with $\epsilon_D = 0.01$, 0.04 and 0.18 from left to right. Top: first-run recovered Stokes I images in Jy/beam, using viridis and a common symmetric logarithmic intensity scale. Bottom: signed $100(I_\epsilon - I_0)/\mathrm{RMS}_{\mathrm{residual},I}$ differences, using bwr and a common linear percentage scale from −30% to +30%. Values beyond these display limits use the endpoint colours, indicated by the extended colourbar. Pixels outside the above-10° mask are excluded. Top-row x axes and y axes in columns 2–3 are omitted.

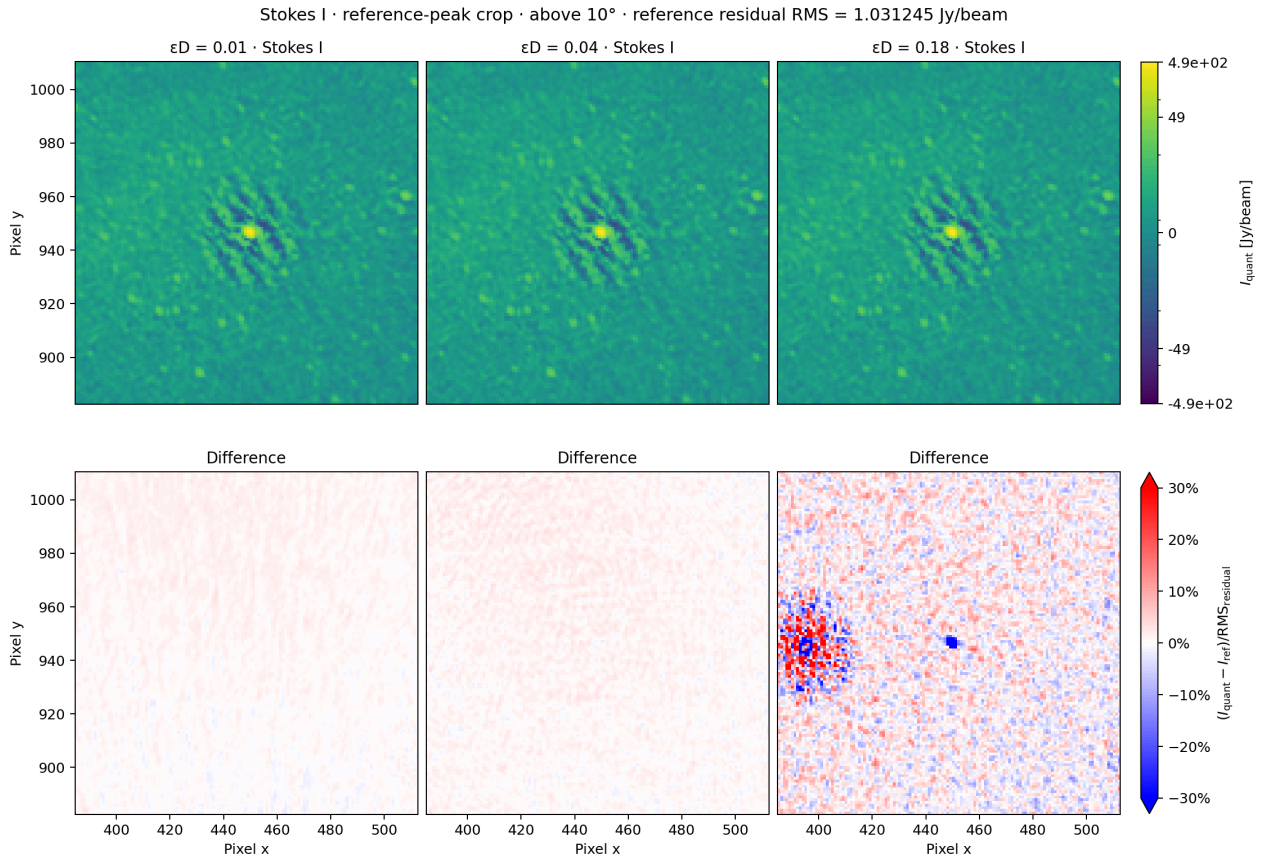


Figure 6. The same two-row, three-column comparisons in a fixed 128×128-pixel crop around the reference peak. Recovered images occupy the top row; percentage differences occupy the bottom row. The full-mask reference residual RMS and both colour scales are identical to Figure 5. The uncompressed reference defines the crop independently of the recovered images.

3.4 Additional Stokes I and V validation

The subsequent test uses the epsilon-D 0.18 archive produced by the measured optimized workflow. Its decoded visibilities equal those of the earlier 0.18 case: 147 column comparisons and the retained metadata match. An independent frozen decoder exports the new archive, and original-source checks pass. Across archive implementations, 5,040 scientific dataset references have matching shapes, dtypes and hashes, with unchanged measured visibility errors. The extra I runs below are distinct CLEAN realizations of the same decoded values; they do not replace the earlier I measurements. In that performance study, V was tested at 0.18; the subsequent three-epsilon V sweep is reported below.

Stokes	Reference residual RMS Jy/beam	Difference mJy/beam, first / repeat	Difference / reference residual RMS %, first / repeat
I	1.031245	49.6019 / 50.2789	4.8099 / 4.8756
V	0.384439	38.6601 / 38.6601	10.0562 / 10.0562

A repeat of each uncompressed reference quantifies observed CLEAN variation:

Stokes	Reference-repeat difference mJy/beam	Difference / reference residual RMS %
I	4.341164	0.420964
V	0.042750	0.011120

All four quantized I/V CLEAN runs reached the final threshold. The V discrepancy is about 10.06% of its reference residual RMS, compared with 4.81–4.88% in I. Although the absolute V difference is smaller, its normalization is also smaller. Both exceed their corresponding reference-repeat differences; the V discrepancy cannot be explained by the observed repeat alone. These controls are not subtracted from the reported errors. Visual agreement or a small I difference does not establish weak-polarization or leakage fidelity. The follow-up below measures the two stricter V cases directly. Measured I/V and identity report.

A follow-up on 27 October images Stokes V at all three epsilon values, using the original verified exports and the same template, reference beam and above-10° mask. Each case is CLEANed twice. All six runs reach the final threshold and preserve their input MS hashes. These new runs are separate from the preceding performance-study V pair at 0.18; no result has been relabelled or substituted. The matching V reference residual RMS is 0.384439444 Jy/beam.

εD	V difference mJy/beam, first / repeat	V difference / reference residual RMS %, first / repeat
0.01	4.1801 / 4.1802	1.0873 / 1.0873
0.04	8.0278 / 8.0278	2.0882 / 2.0882
0.18	38.6601 / 38.6601	10.0562 / 10.0562

Full three-epsilon V measurements. Figures 7–8 use the same layout as the I comparisons in Figures 5–6. The V reference selects a high-elevation crop independently of the reconstructions.

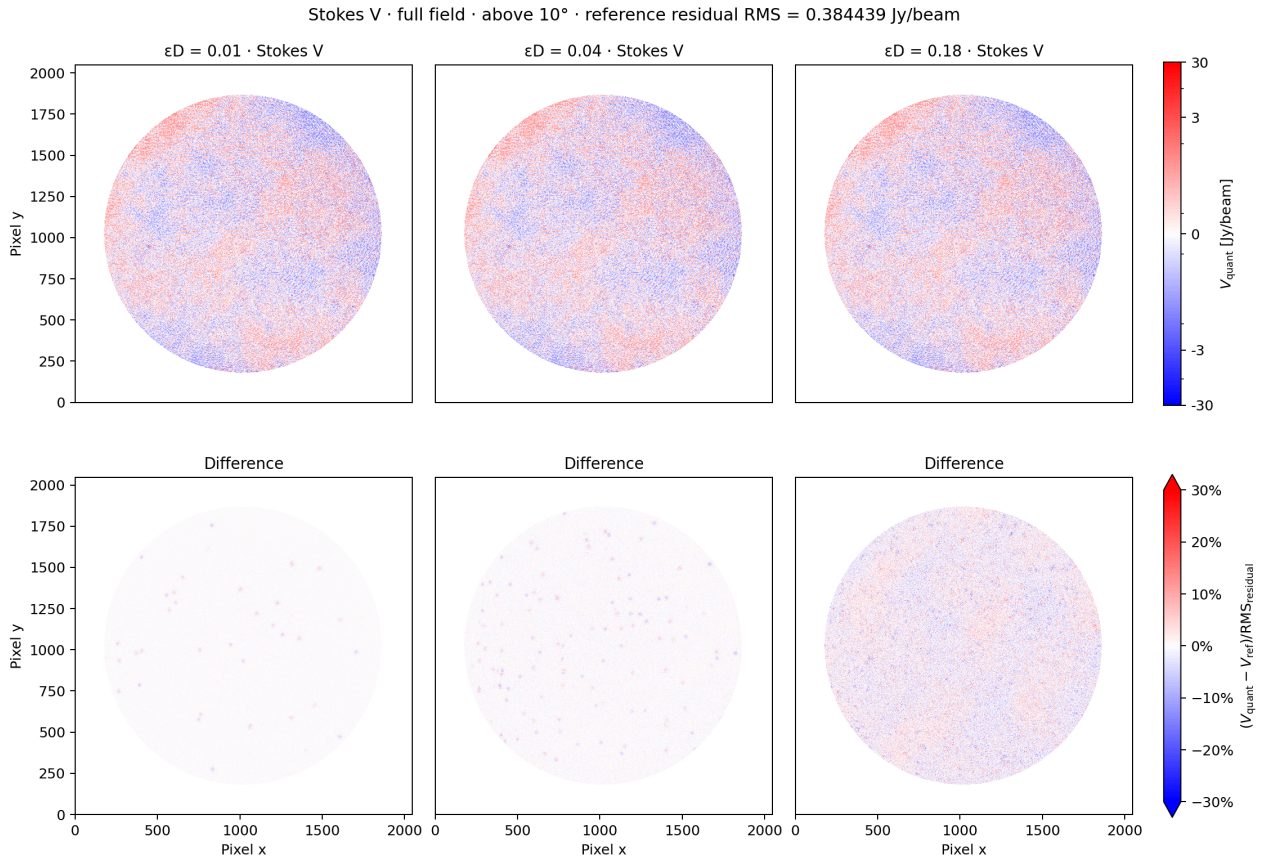


Figure 7. Stokes V counterparts to Figure 5: recovered images above signed percentage differences, with epsilon-D 0.01, 0.04 and 0.18 from left to right. Both rows use bwr; the intensity scale is symmetric logarithmic with fixed limits of −30 to +30 Jy/beam, and the RMS-normalized difference scale is linear from −30% to +30%. Extended difference-colourbar ends indicate values outside that display range. Only the fixed above-10° mask is shown. Top-row x axes and the last two columns' y axes are omitted.

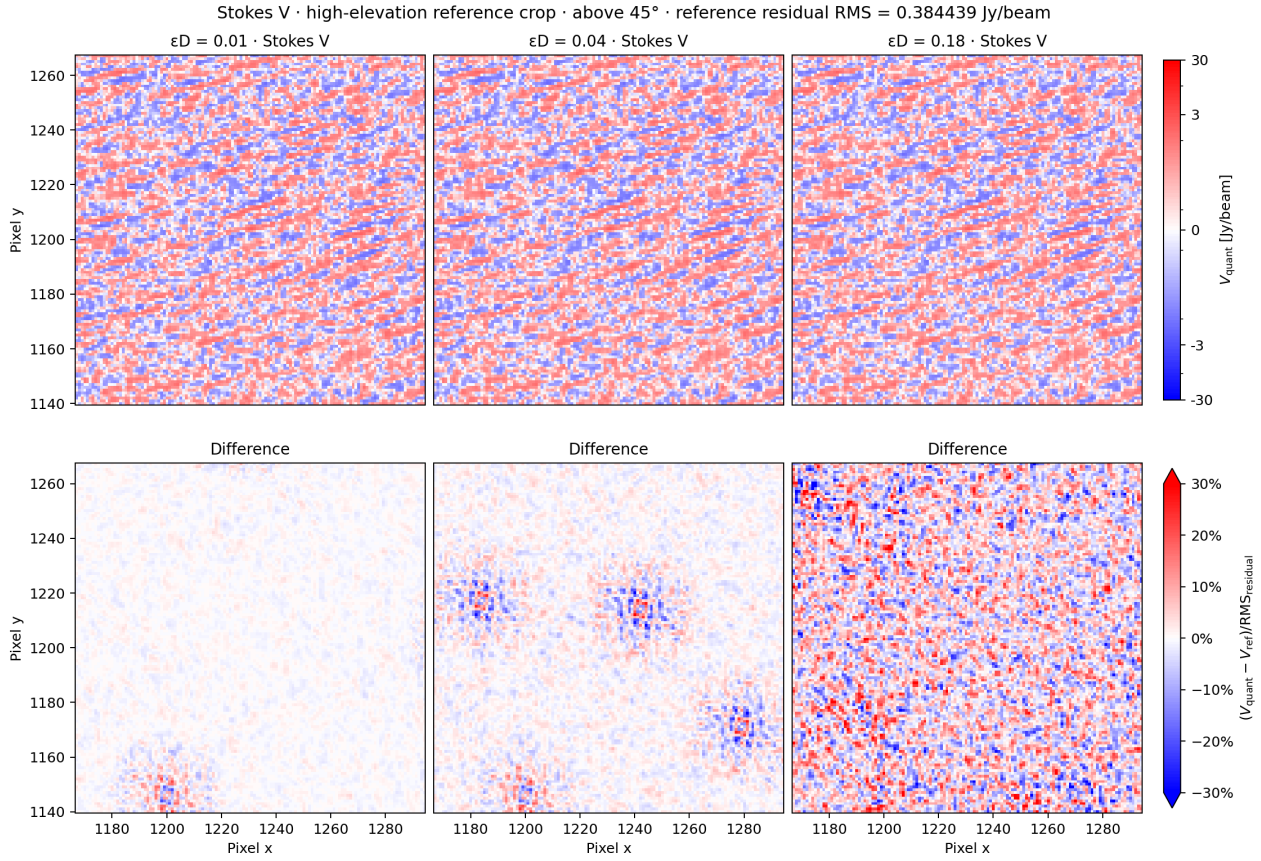


Figure 8. The same V comparisons in a fixed 128×128-pixel crop around the largest absolute V reference pixel among those above 60° elevation at every integration. Every pixel in the crop remains above 45° throughout the observation. The full-field residual RMS denominator and colour scales are retained. The reference defines the crop independently of the reconstructions; the peak is not assigned a source identity.

3.5 Before/after sizes and storage contributors

Representation	Whole MB	Visibility MB	Scope
Original full MS (physical)	4428.204	—	Includes raw and model columns
Original CORRECTED_DATA (logical)	—	1431.429	Denominator for payload fractions
Calibrated-only, exact retained values	985.236	955.400	Shared single-file lossless control
Quantized, εD=0.01	427.429	327.207	Initial three-epsilon study
Quantized, εD=0.04	350.112	249.542	Initial three-epsilon study
Quantized, εD=0.18	242.766	142.163	Initial three-epsilon study
Quantized, εD=0.18, optimized	242.769	142.163	I/V and speed benchmark, run 1

The full-source-to-archive change includes a deliberate preservation-policy change: raw DATA and MODEL_DATA each contain 1431.429 logical MB and are omitted; original flagged visibility values are zeroed. Their combined 2862.858 logical MB must not be interpreted as separately measured physical file savings. The exact control uses the same calibrated-only policy, flags and exact-on-this-dataset four-bit weights, with no visibility quantization.

Its verified shared-file archive occupies 985.236 MB. Before lossless structural sharing, its directory bundle occupied 1079.629 MB; sharing and container compaction save 94.393 MB without requantization.

Relative to this retained-data lossless control, the optimized 0.18 archive is 4.058× smaller (24.641% remaining). Its visibility stream is 9.932% of logical complex64 bytes, or 14.880% of the already losslessly compressed control stream. The complete archive is 5.482% of the physical source, but that ratio combines omission, lossless compression, deduplication and quantization. The 3,630-byte increase relative to the earlier 0.18 file is provenance/container overhead; visibility bytes are unchanged. Lossless sharing measurements, storage audit.

The optimized run-1 archive has the following physical allocation. Every byte is included, including container overhead; percentages may round independently.

Contributor	Bytes	% of archive
Visibility blocks (codes, headers, factors, exceptions, exact blocks)	142,163,290	58.559
Measured visibility-error records	42,153,640	17.364
Quantization/reference statistics	21,032,024	8.663
Original coordinates and identities	19,510,556	8.037
Baseline permutations	6,185,493	2.548
Shared subtables	5,531,017	2.278
Container, manifests, verification and links	5,345,063	2.202
Reconstruction maps	462,886	0.191
Visibility-block indexes	352,749	0.145
Flags and weights	32,166	0.013
Other auxiliary columns	292	0.000
Total	242,769,176	100.000

Visibility payload contributes 58.56% of the file. Error records and quantization statistics together occupy 63.186 MB (26.03%). These diagnostic arrays, followed by coordinates, are the main remaining costs; metadata has not disappeared into a visibility-only compression claim.

The 0.18 precision distribution is shown numerically below. Each width applies to both real and imaginary components; a four-bit component block has eight nominal code bits per complex sample. Figure 4 compares all three epsilon cases.

Bits/component	Blocks	Complex samples	% samples	Stored block MB
3	0	0	0.000	0.000000
4	17,997	110,527,680	61.772	107.846086
5	4,295	26,333,664	14.717	32.098492
6	56	165,216	0.092	0.206102
8	28	172,032	0.096	0.285262
10	0	0	0.000	0.000000
12	0	0	0.000	0.000000
14	0	0	0.000	0.000000
16	0	0	0.000	0.000000
Exact float32	6,848	41,730,048	23.322	1.727348

The 29,224 blocks include 22,376 quantized and 6,848 exact blocks. All exact blocks in this case contain positive-zero placeholders: 41,730,048 samples but only 1.727348 MB after lossless compression. This does not recover discarded flagged values. The remaining 22,564

flagged placeholders are exact exceptions inside mixed blocks. Thus a large exact sample fraction does not imply a large exact-byte cost. Widths unused here remain available to other data.

Before secondary compression, packed integer codes occupy 144.036648 MB, exact-block component arrays 333.840384 MB, exception positions 2.294332 MB and exception values 4.588664 MB. Headers total 0.736096 MB, including 0.358016 MB for the four float32 factors (two centre components, range and gamma). Factor bytes are a subset of headers, not an additional category. These are logical precompression counts: codes and exceptions share compressed block bodies, so their final physical contributions cannot be uniquely separated. The measured combined visibility stream is 142.163290 MB.

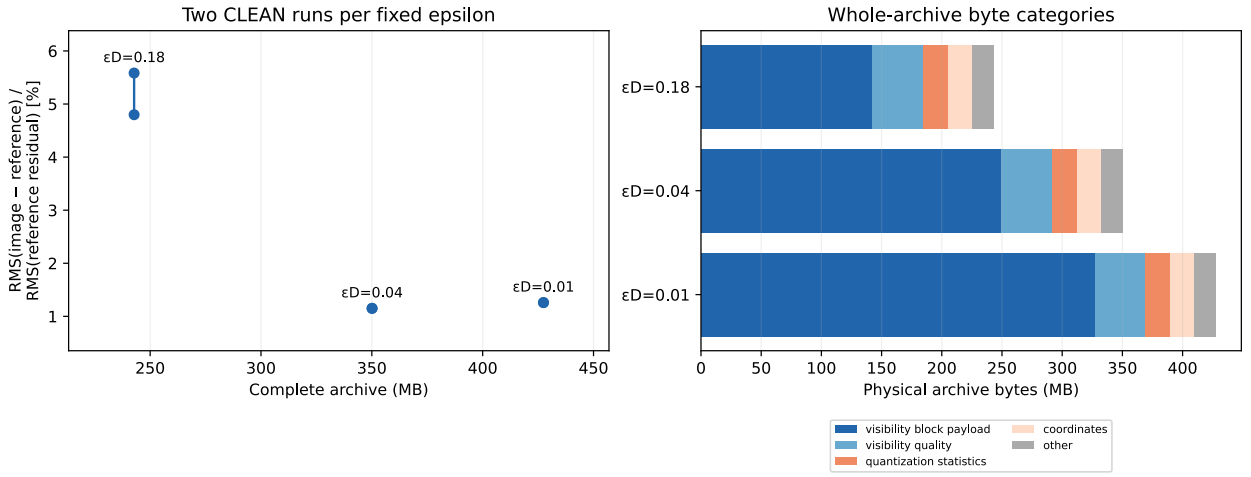


Figure 9. Initial three-epsilon Stokes I study: complete archive size versus RMS(image − reference) / RMS(reference residual), in percent for both CLEAN runs (left), and physical byte contributors (right). The 0.18 visibility payload is identical in the subsequent I/V study. Full optimized-file accounting is given above; nominal component widths alone do not determine whole-archive size.

3.6 Measured MS-to-archive performance

Three unprofiled, fresh-process runs of each implementation measure elapsed wall time from the complete 15-integration MS to a verified single-file archive. The boundary includes discovery, reference sampling, inspection, encoding, integrity and source checks, and final compaction/validation. It excludes MS export and imaging. Scientific settings, flags/weights policy and decoded values are held fixed at epsilon-D 0.18; no error bound is relaxed for this comparison.

The host has two Intel Xeon Silver 4214 CPUs, 48 logical CPUs and 376.5 GiB RAM. Runs use a warm OS cache without eviction. Task-owned timing runs execute sequentially, separately from profiling, unit tests and CLEAN; unrelated activity on this shared host is not controlled. Six one-run process/thread allocation pilots preceded selection of eight processes with six encoder threads each. The table reports the subsequent three repeats, not the fastest pilot.

Execution	Three full-workflow runs, s	Median s	Min–max s
Baseline (1 process × 24 threads)	489.143 / 464.957 / 485.870	485.870	464.957–489.143
Optimized (8 processes × 6 threads)	71.636 / 72.685 / 72.965	72.685	71.636–72.965

The median falls from 485.870 to 72.685 s, a 6.685× speedup. In the optimized repeats, verified-bundle creation takes 53.105–56.132 s and single-file compaction/validation another

16.592–19.325 s. The complete-MS 15-second target remains unmet. Parallel stage-time sums are not elapsed latency. This benchmark compares complete execution configurations; it does not isolate algorithmic changes from the increased worker allocation.

Reported parent peak RSS is 0.991–1.016 GB for the baseline and approximately 0.136 GB for the optimized workflow; the largest individual optimized worker peak is 0.714–0.738 GB. A simultaneous aggregate memory peak was not measured, so these values are not a total-memory comparison. Separately, the frozen independent decoder exported the optimized archive in 58.35 s; export plus source verification and resampling took 118.08 s. That check overlapped later imaging and is not a repeated decoder-performance benchmark.

All repeated archives preserve the same 142,163,290-byte visibility payload, precision distribution and reconstructed values. Their few-hundred-byte total size variation comes from provenance. The performance validation reports 124 passing tests, independent decoding and exact retained-column checks. Detailed profiles, software records and reproduction commands are kept in the benchmark note and measured report.

The following breakdown uses the recorded median run (run 2), whose full process wall time is 72.685 s. Rows are non-overlapping elapsed intervals, or explicit differences of those measured intervals, and sum to the complete run.

Phase	Wall seconds	% of total
Global reference sampling	6.141	8.45
Remaining bundle work: discovery, parallel inspection/ encoding/source checks, finalization	46.964	64.61
Validate input bundle for compaction	6.117	8.42
Copy/share arrays into final archive	4.848	6.67
Validate final archive	5.309	7.30
Other compaction/finalization overhead	3.051	4.20
Process startup/shutdown and remaining orchestration	0.255	0.35
Total process wall time	72.685	100.00

The remaining bundle interval includes discovery, overlapping partition work and finalization. The instrumentation does not separate these into individual additive wall-time contributions. The two overhead rows are measured remainders. Input-bundle validation, array copying/ sharing and final-archive validation are sequential subphases of the 19.325-second compaction stage.

For context, the same run also records the following work summed across the 15 partitions. These totals overlap in wall time and must not be added to the preceding table:

Parallel work	Summed partition seconds
Inspection	23.466
Encoding and writing, including integrity finalization	223.877
Source/integrity verification	73.778

Encoding/writing is the largest recorded partition-work total. The parent waits while partition processes run, so attributing that interval to parent CPU or adding thread/partition durations would misrepresent the 72.685-second latency. A finer additive split inside parallel encoding

would require new instrumentation; none is inferred from unrelated single-context profiles. Median-run timing audit.

4. Conclusion and discussion

Local median subtraction, full-range unsigned power levels and an explicit ϵ_D acceptance rule provide a compact and interpretable encoder policy. The coefficient expresses allowed error relative to block variation, while actual reconstruction checks determine which precision is acceptable. Nine quantized choices plus exact fallback fit within a four-bit precision selector. Broader precision coverage does not remove the need to count factors, exceptions, permutations, indexes, coordinates and error summaries in the archive size.

The three fixed epsilon-D values produce archives of 242.766–427.429 MB; the initial Stokes I study has worst-of-two differences of 1.151–5.584% of reference residual RMS. Epsilon-D 0.01 gives a smaller dirty-image difference than 0.04 (0.146% versus 0.529%), yet a larger worst restored difference (1.260% versus 1.151%). Tighter visibility limits therefore do not impose a monotonic ordering on these CLEAN results. Two repeats do not establish significance or a causal explanation for this reversal.

The additional 0.18 study measures I differences of 4.810–4.876% and V differences of 10.056% of their respective reference residual RMS. V is more affected by this measure than I, and exceeds its observed reference-repeat variation. The 242.769 MB complete archive is 4.058× smaller than the same-policy lossless control; visibility payload is only 58.56% of the final file, with quality and reference-statistics arrays contributing a further 26.03%. Full conversion takes a median 72.685 s, including verification, versus 485.870 s for the baseline. These are measured size, fidelity and speed tradeoffs for one observation; none defines a universal optimum or a scientific acceptance threshold.

A single block statistic cannot establish every future science requirement. Median deviation can vanish for sparse nonconstant data, forcing exact storage; strong and weak products can have different scientific tolerances. Deterministic rounding can create structured errors despite a small mean-square bound. Image agreement depends on visibility weighting and nonlinear deconvolution as well as visibility error, so dirty-image agreement alone is insufficient. These are reasons to retain direct image checks and explicit maximum/bias guards.

The present validation covers joint-frequency Stokes I and V imaging on one supplied calibrated observation. It does not establish polarimetric leakage, narrow spectral-feature, calibration-solution or transient-light-curve fidelity, and contains no held-out observation. Python-casacore reconstruction and WSClean are directly exercised. CASA 5.4.1 reopened the export and produced a 64×64-pixel natural-weight Stokes I dirty-image smoke test on a private copy. Its installed observatory, Earth-orientation and leap-second tables issued warnings; this is not a CASA 6 or precision-astrometry validation. A matched Dysco benchmark, additional observations and dedicated weak-polarization checks remain necessary before choosing general archival defaults. The conservative default remains lossless.

The public repository contains the format, reference decoder, configurations, conformance tests, machine-readable validation and figure-generation scripts. Detailed operational history is retained in development notes and the preserved initial manuscript, rather than being appended to this methods paper.